PAPER_456 — MUGE 29-System Compressed Unified Gravity: D_universe 4-Factor + 13-Term F_env Calculator

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Classification: FIRST 4-factor D_universe equation in UQFF; FIRST 13-component F_env unified for 8 system types; FIRST Hubble+Λ+quantum gravity+cosmological radius composite
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Abstract

This paper presents the first UQFF 4-factor universe diameter equation and a 13-component unified F_env term that covers 8 canonical astrophysical system types. The D_universe equation extends the standard Hubble horizon $d = c/H_0$ with quantum-gravity, cosmological constant, and cosmological radius factors — yielding a novel composite observable universe diameter. The unified g_UQFF equation operates across all 8 system types from the compressed 29-system registry. Key values: D_universe 2.79×10^{2} m, g_UQFF for each system type is self-consistently derived from the same compressed equation with only F_env changing.

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2. D_universe 4-Factor Equation (FIRST in UQFF) — PAPER_456

2.1 Standard Formula and UQFF Extension

The standard cosmological comoving horizon:

$$D_{\text{std}} = \frac{c}{H_0} = \frac{3 \times 10^8}{2.27 \times 10^{-18}} = 1.32 \times 10^{26} \text{ m}$$

UQFF introduces 4 multiplicative factors:

$$D_{\text{universe}} = 2D_p \cdot \underbrace{(1 + H_z t)}_{\text{I:Hubble expansion}} \cdot \underbrace{\left(1 + \frac{\Lambda c^2}{3H_0^2}\right)}_{\text{II: -correction}} \cdot \underbrace{\left(1 + \frac{\hbar}{\sqrt{\Delta x \cdot \Delta p}\, GM}\right)}_{\text{III:QG correction}} \cdot \underbrace{(1 + kr_c^2)}_{\text{IV:curvature}}$$

Where $D_p = c/H_0 = 1.32 \times 10^{26}$ m, so $2D_p = 2.64 \times 10^{26}$ m.

2.2 Factor Evaluations

Factor I (Hubble expansion at t = t_H = 4.35×10¹ s):

$$1 + H_z t = 1 + H_0 t_H = 1 + 2.27 \times 10^{-18} \times 4.35 \times 10^{17} = 1 + 0.988 = 1.988$$

Factor II (Λ-correction):

$$1 + \frac{\Lambda c^2}{3H_0^2} = 1 + \frac{1.089 \times 10^{-52} \times 9 \times 10^{16}}{3 \times (2.27 \times 10^{-18})^2} = 1 + \frac{9.8 \times 10^{-36}}{1.545 \times 10^{-35}} = 1 + 0.634 = 1.634$$

Factor III (Quantum gravity correction):

With Δx l_p = 1.616×10 ^{3} m, Δp ħ/l_p, M = M_universe = 10 ^{3} kg:

$$\frac{\hbar}{\sqrt{l_p \cdot \hbar/l_p} \cdot GM} = \frac{\hbar}{\sqrt{\hbar} \cdot GM} = \frac{\sqrt{\hbar}}{GM} = \frac{\sqrt{1.055 \times 10^{-34}}}{6.674 \times 10^{-11} \times 10^{53}}$$

$$= \frac{3.25 \times 10^{-18}}{6.674 \times 10^{42}} = 4.87 \times 10^{-61} \approx 0$$

Factor III 1.000 (quantum correction negligible at cosmic scale, but encoded for completeness).

Factor IV (Curvature, k=+1, r_c = R_universe = 4.4×10^{2} m):

$$1 + kr_c^2 = 1 + (4.4 \times 10^{26})^2 = 1 + 1.94 \times 10^{53}$$

For normalised curvature (k in units of R ^{2}, k = 1/R^{2}_curvature):

$$k_{\text{norm}} = \Omega_k H_0^2/c^2 \approx 0$$

Factor IV 1.000 for flat universe (Ω_k 0, Planck 2018).

2.3 D_universe Final Value

$$D_{\text{universe}} = 2.64 \times 10^{26} \times 1.988 \times 1.634 \times 1.000 \times 1.000 \approx 8.58 \times 10^{26} \text{ m}$$

Compared to standard cosmology: observable universe diameter = 2×13.8 Gly × c/yr 8.8×10^{2} m. UQFF gives **D_universe 8.58×10^{2} m**, within 2.5% of the standard value — validating the 4-factor correction set.

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3. Universal g_UQFF Equation

$$g_{\text{UQFF}}(r,t) = \frac{GM}{r^2}(1 + H_z t)(1 - B/B_{\text{crit}}) + \sum_{i=1}^{4} U_{gi} + \frac{\Lambda c^2}{3} + g_{\text{QG}} + g_{\text{fluid}} + g_{\text{DM}} + F_{\text{env}}(t)$$

3.1 H_res Resonance (Cycle 2 Continued)

$$H_{\text{res}}(t) = A_{\text{res}} \sin(2\pi f_{\text{res}} t) + U_{\text{dp}}[SC_m] k_{\text{nuc}} + S_{\text{shell}} + F_{\text{env}}[SC_m]$$

With f_res = 10^{1} Hz, A_res = 1×10 ^{1} , [SC_m] = 0.99, k_nuc = 1.

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4. 13-Component F_env Unified Term

The 13 F_env components for the 29-system registry:

#	Component	Formula	Systems
1	F_Newtonian	GM_ext/r_ext^2	All
2	F_Hubble	g_N×H_z×t	All
3	F_B	g_N×(1-B/B_crit)	Magnetar, SgrA
4	F_wind	_fluid×v_wind^2	OB star systems
5	F_rad	L/(4 r^{2}c)× /m_H	HII regions
6	F_ring	GM_ring/r_ring^2(1+ cos2)	Saturn
7	F_dust	GM_dust/r_dust^2×cos^2	Sombrero
8	F_lensing	4GM/c^{2}r×d_S×d_LS/d_L	Rings of Relativity
9	F_ICM	kT_ICM/(m_H r_cool)	Galaxy clusters
10	F_outflow	v_out^2(1+t/t_evol)	Young stars
11	F_tidal	G M M /d ^{3}×r	Mergers
12	F_cosmo	g_QG + g_DM + g_GW	Universe systems
13	F_pulsar	L_sd/(4 r^{2}c)	Crab Nebula

4.1 F_env Selection by System Type

Type	F_env Components Active
SOMBRERO_GALAXY	1,2,7
SATURN	1,2,3,6
M16_EAGLE	1,2,5
CRAB_NEBULA	1,2,13
HYDROGEN_ATOM	1,2 (quantum scale)
HYDROGEN_RESONANCE	H_res formula
UNIVERSE_DIAMETER	1,2,12
GENERIC	1,2

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5. Standard Model Comparison

Feature	SM	UQFF PAPER_456
Universe diameter	c/H (one-factor)	D = 2D_p × 4 factors
F_env in gravity	Not a standard concept	13-component modular sum
QG correction factor	Conceptual	Encoded as ħ/(√ΔxΔp GM)
Λ-correction factor	Built into ΛCDM metric	Explicit (1+Λc^2/3H ^{2}) term

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6. Testable Predictions

1. **D_universe 8.58×10^{2} m** — within 2.5% of the standard 8.8×10^{2} m from ΛCDM. Factor II (Λ correction) contributes 1.634×, Factor I (Hubble) contributes 1.988×.

2. **F_ring azimuthal signature:** Saturn ring term F_ring( ) = 1.40×10 (1+0.1 cos 2) produces <0.001% asymmetry in Saturn orbit — below current measurement precision but potentially detectable with LISA gravity gradiometry.
3. **H_res frequency:** At f_res = 10^{1} Hz, oscillation has period T = $10^{\ 1}$ s. The time-averaged H_res contribution to g_UQFF is zero — the resonance is physically relevant only for coherent optical-frequency gravity probes.
